

DS 3010: Applied Data Modeling and Predictive Analysis (Fall 2025)

Project Proposal

1 Project Title

Your machine Learning project.

- RWIS-ML

2 Team Members

- Harrison Voss
- Alex Mears

3 Project Details

3.1 Project Objective

Please describe your project objective in this part.

- What is the project Objective?
 - The objective of the project is to train a model to identify weather conditions on Iowa roads.
- What problem to solve?
 - Given a set of atmospheric and pavement data, predict the classification of the road condition. i.e: icy, wet, dry.
- Why the problem is important?
 - Driving on roads with adverse conditions is dangerous, so being able to identify the conditions accurately can help drivers be safer.
- Why machine learning can help to solve the problem?
 - The amount of data that is needed to solve the problem is too large for a human to program. Road conditions don't necessarily follow a nice pattern, so we need to use machine learning.

3.2 Datasets

Please describe your dataset in this section.

- What is the data and where you obtain it?
 - The data is archived weather data and webcam images from ISU Mesonet: <https://mesonet.agron.iastate.edu/>
- How the data is collected?
 - The data is available in large quantities from the ISU Mesonet website.

- What will be the features and labels you will use?
 - We will use the features of; ambient temperature, humidity, dew point, pavement temperature, etc. The label is the condition of the road.
- How many examples for training, validation, and testing?
 - In total there are millions of entries for data. From just Jan 1, 2026 to Sept 14th, 2026 there are 3.7 million data points. Once we have built the model, we will do an 80/20 training to initially validate and test it. Throughout the rest of the semester, we will continue to test and validate the model with the real time data.

3.3 Machine Learning Algorithm

Please describe the machine learning algorithm you want to use for your project.

- We will use Logistic Regression.

Please justify your selection.

- After talking to Mobina, we decided on Logistic Regression. This is a classification problem with multiple classes, so Logistic Regression makes the most sense.

3.4 Expected Outcomes

What is your expected outcome for this project?

- To have a model that can predict the condition of a road given atmospheric and pavement data, with an accuracy of at least 90%. And be able to predict the condition of the road between ≈ 30 minutes to ≈ 2 hours in the future with at least 75% accuracy.